# Wen Rui Sun

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University of Alberta
Department of Mathematical and Statistical Sciences

## Research Interests

Convex Geometry, Combinatorial Geometry, Geometric Tomography, Geometric Functional Analysis

## Education

**University of Alberta**, Edmonton, Canada

PhD in Pure Mathematics — 2022—present (expected August 2027)
Supervisors: Vladyslav Yaskin and Beatrice-Helen Vritsiou

MSc in Pure Mathematics — 2020–2022
Thesis: Illumination of Convex Bodies with Symmetries in Dimensions 3 and 4
Supervisors: Vladyslav Yaskin and Beatrice-Helen Vritsiou

BSc in Mathematics (Specialization) — 2018–2020

BSc in Petroleum Engineering — 2013–2018

## Publications

3. **On Axial Symmetry in Convex Bodies**
with *R. Goenka*, *K. Moore*, and *E. P. White*. Pacific J. Math. **341** (2026), no. 2, 275–303.
doi:10.2140/pjm.2026.341.275

2. **Illuminating 1-unconditional convex bodies in $\mathbb{R}^3$ and $\mathbb{R}^4$, and certain cases in higher dimensions**
with *B.-H. Vritsiou*. Canad. J. Math. (2025), 1–67.
doi:10.4153/S0008414X25101260

1. **On the illumination of 1-symmetric convex bodies**
with *B.-H. Vritsiou*. Pure Appl. Funct. Anal. **10** (2025), no. 6, 1483–1524. Issue in honor of Prof. Nicole Tomczak-Jaegermann.
arXiv:2407.10314

## Preprints

1. **On determination of convex bodies by their section functions**
with A. Fish, D. Ryabogin, and V. Yaskin. Preprint, 2026.
arXiv:2610.10229.

## Talks

6. **Illumination of Symmetric Convex Bodies and Some Non-Symmetric Generalizations**
at *Applications of Harmonic Analysis to Convex Geometry*, BIRS, Banff, Alberta, Canada. 2026.

5. **Illumination of 1-Symmetric Convex Bodies**
at *AMS Spring Central Sectional*, NDSU, Fargo, North Dakota, USA. 2026.

4. **Applications of Deep Illumination for the Illumination of 1-Unconditional Bodies**
at *AMS Fall Central Sectional*, Saint Louis University, Saint Louis, Missouri, USA. 2025. (Delivered online).

3. **Measure of Axiality for Convex Figures**
at *CMS Winter Meeting*, Montreal, Quebec, Canada. 2023.

2. **Illumination Conjecture for Convex Bodies with many Symmetries**
at *Harmonic Analysis and Convexity*, BIRS, Banff, Alberta, Canada. 2023.

1. **Illumination Conjecture for Convex Bodies with many Symmetries**
at *Alberta Mathematics Dialogue*, Mount Royal University, Calgary, Alberta, Canada. 2023.

## Poster Presentations

1. **Illumination of 1-Symmetric Convex Bodies**
   at *Recent Advances in Applications of Harmonic Analysis to Convex Geometry*, Kent State University, Kent, Ohio, USA. 2023.

## Awards

**Alberta Graduate Excellence Scholarship**, University of Alberta 2025–2026
**Alberta Graduate Excellence Scholarship**, University of Alberta 2024–2025
**NSERC Undergraduate Student Research Award (USRA)**, University of Alberta 2019–2020
**First Class Standing**, University of Alberta 2018–2019
**First Class Standing**, University of Alberta 2013–2014
**University of Alberta Academic Excellence Scholarship** 2013
**Alexander Rutherford Scholarship** 2010–2013
**International Baccalaureate Certificate** 2013
**Outstanding Honor Student**, Sir Winston Churchill High School 2013

## Teaching Experience

**Course Instructor**, University of Alberta Fall 2025; Spring/Summer 2026
MATH 154 – Calculus for Business and Economics I.
Responsible for lectures, assignments, quizzes, and examinations.

**Graduate Teaching Assistant**, University of Alberta September 2020–Present
Led eleven laboratory sections in MATH 100 (Calculus I), MATH 102 (Applied Linear Algebra), MATH 209 (Calculus III), and MATH 201 (Differential Equations). Also graded assignments and examinations for other mathematics courses.

**Mathematics Tutor**, Decima Robinson Support Centre
Provided one-on-one mathematics assistance, helping students identify mistakes and work through problems independently.